

Software Requirement Specification (SRS) – Hotel Booking System (Enhanced Version)

1. Introduction

1.1 Purpose

This document describes the requirements and features of the Hotel Booking System. The system allows customers to search and book hotels easily, while hotel owners can manage their listings, offers, and reservations efficiently.

1.2 Intended Audience

- **Developers:** To understand system architecture and database design.
- **Project Managers:** To plan and track development progress.
- **Testers:** To validate system functionalities.
- **Hotel Owners:** To understand how to add and manage hotels.
- **Customers:** To use the system for searching and booking hotels.

1.3 Product Scope

- Search and compare hotels by location, price, and rating.
 - Book rooms with options for packages (Full Board / Half Board), number of nights, and number of guests.
 - Manage hotels, rooms, pricing, and availability for hotel owners.
 - Smooth user experience for customers and easy management for hotel owners.
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2. Overall Description

2.1 Product Perspective

The Hotel Booking System is an online **Web/Mobile** platform connecting customers and hotel owners. It has two main interfaces: - **Customer Side:** Search, view details, and book hotels. - **Hotel Owner Side:** Add and manage hotels, rooms, and bookings.

2.2 System Features / Functional Requirements

ID	Feature Description
FR1	Search hotels by location, price, rating, and room availability.
FR2	View hotel details, images, amenities, and available room types.

ID	Feature Description
FR3	Manage bookings: create, update, cancel.
FR4	Manage hotels and rooms: add hotels/rooms, update prices and offers.
FR5	Online payment and issue booking receipts.
FR6	Customer reviews and ratings for hotels.
FR7	Notifications for new bookings or updates.

2.3 User Classes and Characteristics

Customer: - Search hotels, view details, and book rooms. - Choose package type and specify guests/nights. - Manage bookings and online payment history.

Hotel Owner: - Add and update hotels, rooms, prices, and offers. - Manage bookings and view reviews. - Update room availability in real-time.

2.4 Operating Environment

- **Hardware:** Any internet-connected device (PC, tablet, smartphone).
- **Software:**
 - Web browser or mobile OS (Android/iOS).
 - Backend: Django.
 - Database: MySQL.

2.5 Constraints

- Secure handling of customer and hotel data.
- Fast and reliable booking process (<3 seconds).
- User-friendly and responsive interface.
- Up-to-date availability and pricing.

3. Non-Functional Requirements

Category	Requirement
Usability	Interface should be simple and intuitive for all users.
Reliability	System must maintain high uptime and data accuracy.
Performance	Searches and bookings respond within 3 seconds.
Availability	Accessible 24/7 on web and mobile platforms.

Category	Requirement
Security	Encrypt all data and protect against unauthorized access.
Maintainability	Easy to update hotel listings and system components.

4. External Interface Requirements

4.1 User Interfaces

- Search pages, hotel detail pages, booking/payment pages, hotel management pages.

4.2 System Interfaces

- APIs for payment gateways, email notifications, and push notifications.

4.3 Communication Interfaces

- Secure HTTPS connection between client and server.

5. Data Requirements

- **Customer:** CustomerID, Name, Email, Password, BookingHistory
- **HotelOwner:** OwnerID, Name, Email, Password, Hotels
- **Hotel:** HotelID, Name, Location, Rooms, OwnerID, Rating
- **Room:** RoomID, HotelID, Type, Price, Availability
- **Booking:** BookingID, CustomerID, RoomID, StartDate, EndDate, Status, PaymentID
- **Payment:** PaymentID, BookingID, Amount, Date, Status
- **Review:** ReviewID, CustomerID, HotelID, Rating, Comment

6. Use Cases (Examples)

1. **Search Hotel:** Customer searches and selects a hotel.
2. **Book Room:** Customer books a room and pays online.
3. **Manage Hotel:** HotelOwner adds or edits hotels and rooms.
4. **Write Review:** Customer leaves a review or rating for a hotel.

7. Performance, Security, and Scalability

- Booking and search response time < 3 seconds.

- AES encryption for passwords and payment data.
- System designed to scale for more users and hotel listings in the future.

8. Use Case Scenarios

8.1 Search for Hotels

Use Case Name: Search Hotels

Primary Actor: Customer

Goal: Find available hotels based on specific criteria.

Main Flow:

1. Customer opens the app/website.
2. Customer enters search criteria (location, date, price range).
3. System searches the database.
4. System displays matching hotels.
5. Customer views hotel details.

8.2 Search for Hotels

Use Case Name: Book Room

Primary Actor: Customer

Goal: Complete a hotel booking.

Main Flow:

1. Customer selects a hotel.
2. Customer selects room type.
3. System checks availability.
4. Customer enters booking details.
5. Customer proceeds to payment.
6. Payment is confirmed.
7. System creates a booking record.
8. System sends confirmation to Customer.

8.3 Manage Hotel (Owner)

Use Case Name: Manage Hotels

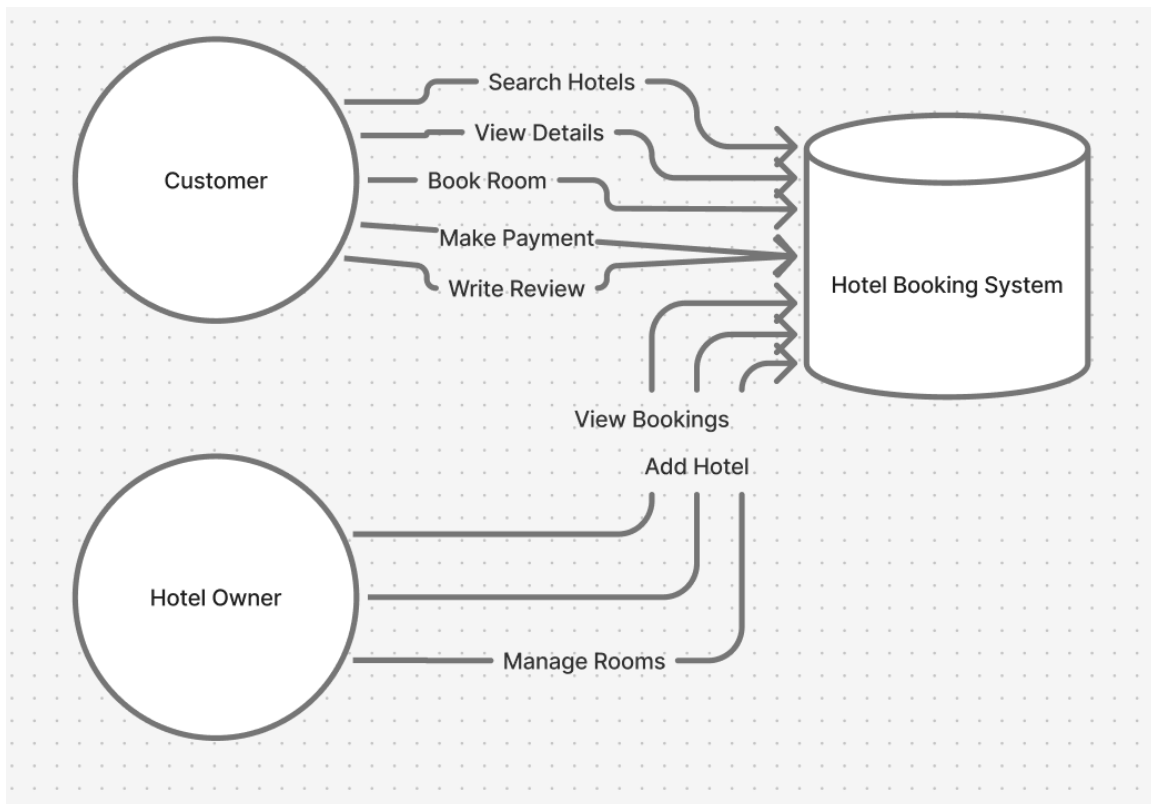
Primary Actor: Hotel Owner

Goal: Add and update hotel details.

Main Flow:

1. Hotel Owner logs in.
2. Owner selects “Add Hotel”.
3. Owner enters hotel information.
4. Owner adds rooms, prices, packages.
5. System saves the data.
6. System displays updated hotel dashboard.

9. Use Case Diagram



9. Sequence Diagram

9.1 Search for Hotel :

Actors: Customer → System → Hotel Database

Sequence Steps:

1. **Customer** enters search criteria (location, date, price range).
2. **System** receives the search request.
3. **System** sends query to **Hotel Database**.
4. **Hotel Database** returns list of available hotels.
5. **System** filters and sorts results.
6. **System** displays search results to **Customer**.

9.2 Book Room:

Actors: Customer → System → Hotel Database → Payment Gateway

Sequence Steps:

1. **Customer** selects a hotel and room type.
2. **System** retrieves room details from **Hotel Database**.
3. **Customer** confirms booking details.
4. **System** sends payment request to **Payment Gateway**.
5. **Payment Gateway** processes payment and returns success/failure.
6. **System** creates booking record in **Hotel Database**.
7. **System** displays booking confirmation to **Customer**

9.3 Manage Hotel (Hotel Owner):

Actors: Hotel Owner → System → Hotel Database

Sequence Steps:

1. **Hotel Owner** logs into the system.
2. **Hotel Owner** selects "Manage Hotels/Rooms".
3. **System** fetches hotel list from **Hotel Database**.
4. **Hotel Owner** adds/updates hotel or room information.
5. **System** sends update request to **Hotel Database**.
6. **Hotel Database** updates stored information.
7. **System** displays success confirmation to **Hotel Owner**.